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Department of Mathematics
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Math 208 Major Exam I Term 242

Original exam date: 18 February 2025. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. A general solution of the exact differential equation

$$(1 + ye^{xy})\,dx + (2y + xe^{xy})\,dy = 0$$

(where c is constant) is

(a) $x^2 + e^{xy} + y^2 = c$

(b) $e^{xy} + y = c$

(c) $x + e^{xy} = c$

(d) $x + e^{xy} + y = c$

(e) $x + e^{xy} + y^2 = c$

2. If the acceleration of a particle is $a(t) = \dfrac{1}{(t+1)^3}$, the initial velocity is $v_0 = 0$, and the initial position is $x_0 = 0$, then $x(1) =$

(a) 1

(b) $\dfrac{1}{16}$

(c) $\dfrac{1}{2}$

(d) $\dfrac{1}{8}$

(e) $\dfrac{1}{4}$

3. If the vectors $u = (3, -4, 5)$, $v = (1, -2, 3)$, and $w = (6, -8, a)$ are linearly dependent, then $a =$

(a) 7

(b) 9

(c) 8

(d) 11

(e) 10

4. The solution of the initial-value problem

$$\frac{dy}{dx} = y^2 \sin x - y^2 \cos x, \qquad y\left(\frac{\pi}{2}\right) = \frac{1}{3}$$

is

(a) $y = \dfrac{1}{\sin x + \cos x + 2}$

(b) $y = \dfrac{1}{\sin x + \cos x}$

(c) $y = \dfrac{1}{-\sin x + \cos x + 4}$

(d) $y = \dfrac{1}{\sin x + \cos x - 1}$

(e) $y = \dfrac{1}{\sin x - \cos x + 2}$

5. Given that $y = \dfrac{1}{c - x}$ is a family of solutions of the differential equation $\dfrac{dy}{dx} = y^2$. The largest interval of existence of the solution for the initial-value problem

$$\begin{cases} \dfrac{dy}{dx} = y^2 \\ y(1) = 2 \end{cases}$$

is

(a) $(-\infty, 3)$

(b) $(-\infty, 2)$

(c) $\left(-\dfrac{3}{2}, \infty\right)$

(d) $\left(-\infty, \dfrac{3}{2}\right)$

(e) $\left(\dfrac{3}{2}, \infty\right)$

6. By using a suitable substitution, we can transform the differential equation

$$3xy^2y' = 3x^4 + y^3$$

into the linear differential equation

(a) $v' - \frac{1}{x}\,v = x^3$

(b) $v' - \frac{1}{x}\,v = 3x^3$

(c) $v' + \frac{1}{x}\,v = x^2$

(d) $v' + \frac{1}{x}\,v = 3x^3$

(e) $v' + \frac{1}{x}\,v = 3x^2$

7. The general solution of the differential equation

$$2xy' - 3y = 9x^3$$

is (where c is a constant)

(a) $y = 3x^3 + cx^{3/2}$

(b) $y = x^3 + cx^{3/2}$

(c) $y = 2x^3 + cx^{3/2}$

(d) $y = 2x^3 + cx^{2/3}$

(e) $y = 3x^3 + cx^{2/3}$

8. A general solution of the homogeneous differential equation

$$2xy\frac{dy}{dx} = 4x^2 + 3y^2$$

is (where c is a constant)

(a) $y^2 - 4x^2 = cx^3$

(b) $y^2 + 4x^2 = cx^3$

(c) $y^2 + 2x^2 = cx^3$

(d) $y^2 - 2x^2 = cx^3$

(e) $y^2 + 4x^2 = cx$

9. The solution space of the system

$$\begin{aligned} x_1 - 3x_2 - 7x_3 - 10x_4 &= 0 \\ x_1 + x_2 + 9x_3 + 2x_4 &= 0 \\ x_1 - 2x_2 - 3x_3 - 7x_4 &= 0 \end{aligned}$$

is the set of all linear combinations of the form $s\mathbf{u} + t\mathbf{v}$, where s and t are real numbers and

(a) $\mathbf{u} = (-5, -4, 1, 0),\ \mathbf{v} = (1, 3, 0, 1)$

(b) $\mathbf{u} = (-5, -4, 1, 0),\ \mathbf{v} = (1, -3, 0, 1)$

(c) $\mathbf{u} = (-5, -4, 1, 0),\ \mathbf{v} = (1, -3, 0, -1)$

(d) $\mathbf{u} = (-5, 4, 1, 0),\ \mathbf{v} = (1, -3, 0, 1)$

(e) $\mathbf{u} = (5, -4, 1, 0),\ \mathbf{v} = (1, -3, 0, 1)$

10. Let $u = (2, 7)$, $v = (3, 8)$ and $t = (0, 5)$. If $t = au + bv$, then $a + b =$

(a) -2

(b) -1

(c) 0

(d) 1

(e) 2

11. Which one of the following subsets is a subspace of $\mathbb{R}^4$?

(a) The set of all vectors (x_1, x_2, x_3, x_4) such that all x_i's are nonzero

(b) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 + x_2 + x_3 + x_4 = 5$

(c) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 = 3x_3$ and $x_2 = 4x_4$

(d) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1x_2x_3x_4 = 0$

(e) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1x_2 = x_3x_4$

12. The general solution of the differential equation

$$x^2y'' + 2xy' = 12x^3$$

is $y =$ (A and B are real numbers)

(a) $x^2 + \dfrac{A}{x} + B$

(b) $x^3 - \dfrac{A}{x} + B$

(c) $x^3 - \dfrac{A}{x} + Bx$

(d) $x^2 - \dfrac{A}{x^2} + B$

(e) $x^3 - \dfrac{A}{x^2} + B$

13. If the differential equation

$$\left(6x^2 + kxy\right)dx + \left(x^2 - 6y\right)dy = 0$$

is an exact differential equation, then $k =$

(a) -1

(b) 0

(c) 2

(d) -2

(e) 1

14. If we substitute $y = e^{rx}$ into the differential equation

$$5y'' - 6y' - y = 0,$$

then the product of all values of r is

(a) $\dfrac{-1}{6}$

(b) 0

(c) $\dfrac{1}{5}$

(d) $\dfrac{-1}{5}$

(e) $\dfrac{1}{6}$

15. When sugar is dissolved in water, the amount A that remains undissolved after t minutes satisfies the differential equation $\dfrac{dA}{dt} = -kA$ $(k > 0)$. If 25% of the sugar dissolves after 1 minute, how long does it take for half of the sugar to dissolve (in minutes)?

(a) $\dfrac{\ln 2}{\ln(4/3)}$

(b) $\ln(4/3)$

(c) $\dfrac{1}{\ln(7/3)}$

(d) $\dfrac{-1}{\ln(3/4)}$

(e) $\dfrac{\ln 2}{\ln(7/4)}$

Answer Key

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Q	Ans	Brief explanation
1	**(e)**	Exact: $F = x + e^{xy} + g(y)$, $g' = 2y$, so $x + e^{xy} + y^2 = c$.
2	**(e)**	$v = \frac{1}{2} - \frac{1}{2(t+1)^2}$, $x = \frac{t}{2} + \frac{1}{2(t+1)} - \frac{1}{2}$, so $x(1) = \frac{1}{4}$.
3	**(e)**	$\det = 20 - 2a = 0$, so $a = 10$ (then $w = 2u$).
4	**(a)**	Separable: $-\frac{1}{y} = -\cos x - \sin x - 2$, so $y = \frac{1}{\sin x + \cos x + 2}$.
5	**(d)**	$c = \frac{3}{2}$, $y = \frac{1}{3/2 - x}$ blows up at $x = \frac{3}{2}$; interval containing 1: $(-\infty, \frac{3}{2})$.
6	**(b)**	Bernoulli $n = -2$, $v = y^3$: $v' - \frac{1}{x}v = 3x^3$.
7	**(a)**	$\rho = x^{-3/2}$: $(x^{-3/2}y)' = \frac{9}{2}x^{1/2}$, so $y = 3x^3 + cx^{3/2}$.
8	**(b)**	$v = y/x$: $xv' = \frac{4+v^2}{2v}$, $4 + v^2 = cx$, so $y^2 + 4x^2 = cx^3$.
9	**(b)**	Row reduce: $x_2 = -4s - 3t$, $x_1 = -5s + t$, so $\mathbf{u} = (-5, -4, 1, 0)$, $\mathbf{v} = (1, -3, 0, 1)$.
10	**(d)**	$2a + 3b = 0$, $7a + 8b = 5$: $a = 3$, $b = -2$, so $a + b = 1$.
11	**(c)**	Only the homogeneous linear system $x_1 = 3x_3,\ x_2 = 4x_4$ gives a subspace.
12	**(b)**	$p = y'$: $(x^2 p)' = 12x^3$, $p = 3x^2 + \frac{A}{x^2}$, so $y = x^3 - \frac{A}{x} + B$.
13	**(c)**	$M_y = kx$, $N_x = 2x$, so exact when $k = 2$.
14	**(d)**	$5r^2 - 6r - 1 = 0$, $r = \frac{3 \pm \sqrt{14}}{5}$, product $\frac{9-14}{25} = -\frac{1}{5}$.
15	**(a)**	$A = A_0 e^{-kt}$, $e^{-k} = \frac{3}{4}$ so $k = \ln \frac{4}{3}$; $t = \frac{\ln 2}{\ln(4/3)}$.